

Experiment 6

CHARACTERISTICS OF A MICROWAVE AMPLIFIER

Experiment 6

Objective:

- To study the small signal S-parameters of an amplifier.
- To understand different classes of amplifiers.
- To study the linearity and power added efficiencies of amplifiers.

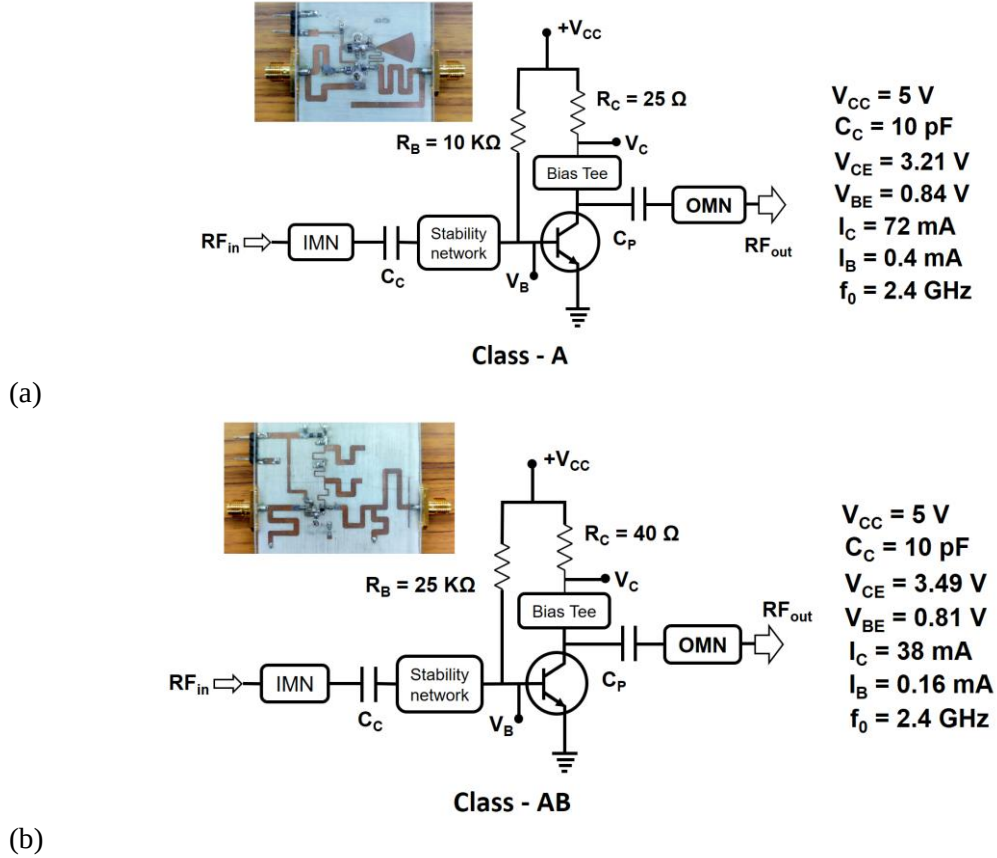


Fig. 1 Photographs of a (a) class A linear amplifier designed for maximum gain @ 2.4 GHz, and (b) a class AB power amplifier (conduction angle = 270°) using the same transistor.

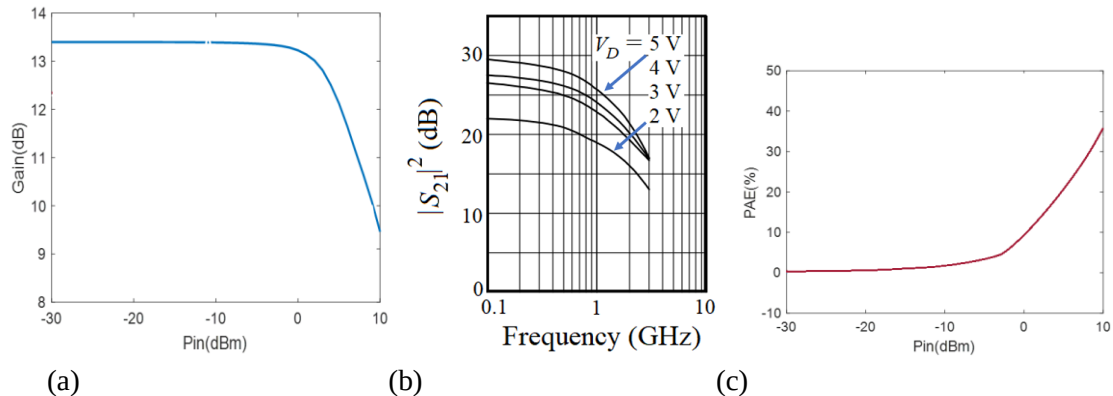


Fig. 2 Measured (a) gain vs. input RF power (class A), (b) low power gain, and (c) PAE vs. input RF power of a class A amplifier implemented for maximum gain at 2.4 GHz.

Theory:

An RF amplifier is used to amplify the input RF signal. Fig.1 shows two RF amplifiers operating in Class A and Class AB modes. The first amplifier is designed for maximum gain, while the second amplifier uses the same transistor but is optimized for maximum efficiency.

The required characteristics of an RF amplifier vary widely depending on its application. In an RF receiver chain, the first component after the antenna must amplify a very weak signal without significantly degrading the signal-to-noise ratio. An amplifier increases the power of both the signal and the noise present at its input. To minimize additional noise and maintain good overall noise performance in the receiver chain (measured as the noise figure), the first component, known as the low noise amplifier (LNA), is designed to minimize noise generation. LNAs typically operate in Class A, with the transistor always on.

On the other hand, in an RF transmitter, the last component before the antenna must generate high power without significant signal distortion. As a result, this amplifier operates at the 1 dB compression point, the universally recognized boundary between linear and nonlinear operation. This type of amplifier is called a power amplifier. RF designers aim to extract the maximum possible RF power without damaging the device, making the power amplifier the component with the highest DC power dissipation in any RF system. To minimize heat generation and DC power consumption, the DC-to-RF power conversion efficiency, measured as power-added efficiency (PAE), must be maximized. PAE varies with amplifier classes and input RF power. Higher classes, like Class C and Class F operations, provide higher PAE but at the cost of greater nonlinearity, as the transistor remains off for most of the RF cycle. A resonant circuit then recovers the other part of the RF sinusoidal signal. There is a common trade-off in amplifier design between efficiency and linearity.

Sometimes, additional linear RF amplifiers, typically designed for maximum gain and operating in Class A, are used as gain blocks. Fig. 2 shows the measured characteristics of an amplifier operating in Class A mode. During the implementation of an RF amplifier, the overall circuit must be checked for stability. Otherwise, the amplifier may become an oscillator, resulting in much lower gain than expected and generating one or more arbitrary RF frequencies.

The design approach at GHz frequencies is different from that used at lower frequencies, because it always emphasizes the concept of maximum power transfer to maximize the signal-to-noise ratio. The input matching network (IMN) is designed to transfer maximum power from a $50\ \Omega$ RF source to the device, while the output matching network (OMN) maximizes power transfer from the device to the $50\ \Omega$ load. Coupling capacitors (C_C) are used to block DC. Otherwise, the DC bias point might shift. A Bias Tee (low pass or band stop filter) is used at the collector terminal to allow DC while blocking RF. This is necessary because RF power cannot be wasted in the R_C resistor or the DC source. Also, the input impedance of the DC source at GHz frequencies is unknown.

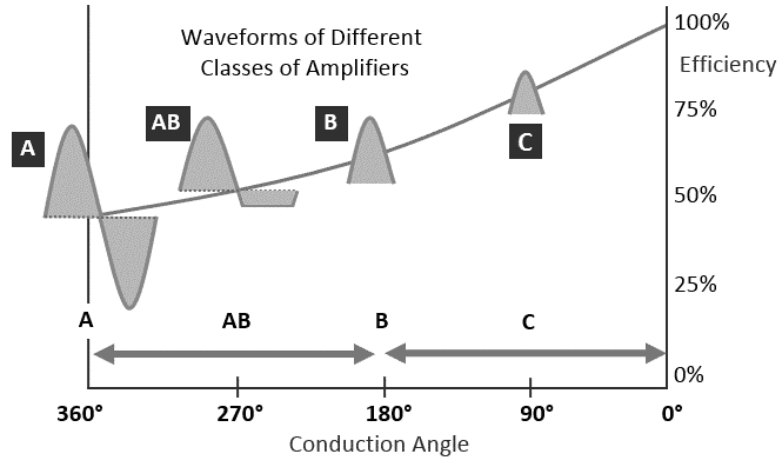


Fig. 3 Conduction angle and efficiency of various classes of amplifiers.

Different classes of amplifiers:

RF amplifiers are divided into several classes, based on how much of the cycle of the sinusoidal signal the transistor is conducting current. Most popular classes are class A, and class AB, which are considered the linear amplifier classes in which the active device is used as a controlled current source while class C is a nonlinear class in which the active device is used almost like a switch. The bias at the input of the active device determines the class of the amplifier. Fig. 3 shows how the PAE varies with the conduction angle.

Noise and Noise Figure (NF):

Noise power is a result of random processes such as the voltage generated between the terminals of a device or component due to the random motions of charges or charge carriers in the device. All components in a microwave system generate noise. Noise also enters from external source. Among different types of noise, the thermal noise due to the thermal vibration of bound charges at a temperature above absolute zero is the most basic source of noise. The noise level of a system sets the lower limit on the strength of a signal that can be detected in the presence of the noise.

The thermal noise is modeled with a Thevenin equivalent circuit consisting of a noiseless resistor R and a generator with a *rms* voltage $v_n = \sqrt{4kTB}$, where $k = 1.380 \times 10^{-23}$ J/K is the Boltzmann's constant, T = the temperature in degrees kelvin (K), B = the bandwidth of the system in Hz. Then, maximum power delivered from the source to a matched load is kTB , independent of the resistance value. An antenna when connected to a LNA delivers signal as well as noise. Noise is also generated inside the LNA. Thus, input SNR of a LNA is more than the output SNR.

Noise figure (NF) and noise factor are measures of degradation of the SNR caused by a component. The noise factor is defined as the ratio of actual output noise to that which would remain if the device itself did not introduce noise, i.e.

$$\text{Noise factor } (F) = \frac{\text{Input SNR}}{\text{Output SNR}}.$$

While measuring the input SNR, the input noise is attributed to thermal noise produced at a standard noise temperature $T_0 = 290$ K. Noise figure is the noise factor in decibel scale i.e. Noise figure (NF) = $10 \log_{10}(F) = SNR_{in}(dB) - SNR_{out}(dB)$.

Power added Efficiency (PAE):

The PAE is defined as the ratio of RF output power with addition of RF input power to the DC input power of a power amplifier. It denotes the percentage of DC power converted to effective AC power by an amplifier. As a mathematical expression, it can be defined as

$$PAE = (P_{RFout} - P_{RFin}) / P_{DC},$$

where P_{DC} is the DC power consumed by the amplifier.

P1dB:

P1dB or 1 dB compression point is a measure of a device's ability to withstand and maintain a certain level of signal power without distortion. It is defined as the output RF power level at which the gain decreases 1 dB from its constant value or small signal value. Beyond this point, any increase in input power causes a greater reduction in output power. Devices with a higher P1dB are generally considered to be better suited for high-power applications. Above this point, the power lost in the harmonics increases, and the waveform deviates significantly from the input waveform shape.

Know your amplifier:

Absolute maximum rating of the transistor BFP650:

- Device current $I_{CC} = 150$ mA, $I_{BB} = 10$ mA
- Device voltage $V_{CE} (+V) = 4$ V
- Total power dissipation at $T_S = 120^\circ\text{C}$, $P_{tot} = 500$ mW.
- Junction temperature $T_j = 150^\circ\text{C}$.

DC operating points for the implemented amplifiers @ 2.4 GHz:

- Class A mode: $I_{CC} = 70$ mA, $I_{BB} = 400$ μA , and $V_{CEQ} = 2.9$ V.
- Class AB mode: $I_{CC} = 40$ mA, $I_{BB} = 170$ μA , and $V_{CEQ} = 3.4$ V.

Procedure:

Identify the transistor, input and output matching circuits, input and output coupling capacitors, the bias Tee and the resistor R_C connected between the bias Tee and the bias point V_{CC} . Connect the circuit and apply a biasing voltage of $V_{CC} = 5$ V. Note that, V_{CE} is less than the absolute maximum value. Wait for 10-15 minutes before turning ON the RF source.

Scattering parameters:

1. Set the VNA power level to -20 dBm or “low level” (**Caution:** An amplifier provides gain. Therefore, $|S_{21}|$ dB is positive. The VNA may be damaged if you don't set the VNA power to a low level).
2. Calibrate the VNA over 2.0-3.0 GHz.
3. Connect the amplifier to VNA.
4. Switch on D.C. supply of the amplifier and wait for 10 min.
5. Measure and plot the S-parameters ($|S_{11}|$, $|S_{22}|$, $|S_{21}|$, and $\angle S_{21}$) over 2.0-3.0 GHz.

P1 dB compression point and PAE:

1. Connect the amplifier input to a RF source.
2. Connect the amplifier output to a Spectrum Analyzer.
3. Set the signal frequency at 2.4 GHz or as suggested.
4. Switch on the amplifier D.C. power supply and wait for 10 min.
5. In the Spectrum Analyzer, set an auto attenuation of 20 dB.
6. Vary the input power level P_{RFin} from -80 dBm to +10 dBm or as suggested. Note down both input and output powers for 10-15 steps.
7. Use an external attenuator at the RF output port of the amplifier when the input power is more than 0 dBm (1 mW). This is to protect the Spectrum Analyzer. Measure the attenuation offered by the attenuator by noting down the difference in RF power level in the Spectrum Analyzer with and without the attenuator.
8. Also using a multimeter, measure and note down the DC voltage drop V_R across R_C for each and every value of P_{RFin} . Then, $V_{CEQ} = V_{CC} - V_R$, and $I_{CQ} = V_R / R_C$. Now, use the relationship for the PAE. Note that the DC power dissipation $I_{CQ} \times V_{CEQ}$ does not vary with the RF cycle but the P_{RFin} . Calculate PAE from above relationship and plot PAE vs. P_{RFin} (**Caution:** Always use a lumped resistor in the range of 1–10 k Ω in series with the multimeter probe, as the multimeter may provide a DC voltage in some modes that could damage the device).
9. Plot the gain variation with P_{RFin} and compare the results with the gain obtained from previous step. (**Caution:** Do not increase the input power level unnecessarily, otherwise you may burn the device).